

NORMALIZED HESSIAN SPACES AND KRONECKER PRODUCTS OF SYMMETRIC PERSISTENT TENSORS

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ABSTRACT. We revisit the Kronecker-closure claim for symmetric persistent tensors in Gharahi’s recent preprint *Kronecker Products of Symmetric Persistent Tensors*. The proof of its polarized Hessian identity establishes the desired perfect-power determinant identity on tuples of decomposable polarization vectors and then extends the proposed multilinear root by the universal property of the tensor product. This does not by itself extend the determinant identity to arbitrary polarization tensors: the obstruction is precisely the Segre ideal already noted earlier in that paper.

No counterexample to the closure theorem is produced here. Instead we develop a normalized polarized-Hessian space that gives several replacement results. If one factor has a simultaneously strictly triangularizable normalized Hessian space and the other has a nil normalized Hessian space, then their Kronecker product is persistent. If, in addition, the certificate of the first factor factors through a single linear form, no nil-space hypothesis is needed on the second factor. A trace-zero Jordan-closure criterion implies the required simultaneous triangularizability by Jacobson’s generalized Engel theorem for weakly closed sets. These arguments give unconditional Kronecker closure when one factor has dimension at most three, in every degree, and for cubics when one factor has dimension at most four.

We also prove a Veronese-contraction lemma reducing the fixed-linear-form condition in arbitrary degree, for each fixed dimension, to a quartic base case. Finally, exact symbolic computations on several hundred persistent examples show Jordan closure with no observed failure, including tests on Kronecker products formed in the full product tensor space. The general implication from persistence to the required normalized-Hessian structure remains open.

1. INTRODUCTION

Persistent tensors were introduced as a class adapted to the substitution method and tensor-rank lower bounds; in the symmetric setting they admit a

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strong Hessian characterization [2, 3]. Gharahi's recent preprint [1] proposes an especially natural closure property: if

$$f \in \operatorname{Sym}^n V, \quad g \in \operatorname{Sym}^n W$$

are symmetric persistent tensors, then their Kronecker product $f \boxtimes g \in \operatorname{Sym}^n(V \otimes W)$ is again persistent.

The purpose of this note is constructive. We first isolate a logical issue in the proof of the polarized Hessian identity in [1, Proposition 10]. The issue does not supply a counterexample, and exact symbolic tests continue to support the stated closure theorem. We then give a matrix-space mechanism that proves the theorem under explicit structural hypotheses and yields several unconditional subcases.

Let $r = n - 2$. By the Hessian characterization of Gharahi–Ottaviani, a persistent form $f \in \operatorname{Sym}^n V$, $\dim V = d$, admits a nonzero multilinear certificate

$$P_f : V^r \longrightarrow \mathbb{C}$$

such that

$$(1) \quad \det H_f(A) = P_f(A)^d, \quad A \in V^r,$$

where $H_f(A)$ is the constant Hessian matrix of the quadratic partial polarization of f at the tuple A . After choosing an admissible base tuple B with $P_f(B) \neq 0$, define

$$\begin{aligned} M_f(A) &= H_f(B)^{-1} H_f(A), & \lambda_f(A) &= \frac{P_f(A)}{P_f(B)}, \\ N_f(A) &= M_f(A) - \lambda_f(A)I. \end{aligned}$$

The central object of this note is the normalized Hessian space

$$\mathcal{N}_f(B) = \operatorname{Span}\{N_f(A) : A \in V^r\} \subseteq \operatorname{End}(V).$$

The principal sufficient conditions are:

- (T) $\mathcal{N}_f(B)$ is simultaneously strictly triangularizable;
- (Nil) every element of $\mathcal{N}_f(B)$ is nilpotent;
- (TR) every element of $\mathcal{N}_f(B)$ has trace zero;
- (JC) $XY + YX \in \mathcal{N}_f(B)$ for all $X, Y \in \mathcal{N}_f(B)$.

We make no general base-independence claim for (Nil), (TR), or (JC); the sufficient conditions below are existential in the choice of an admissible base.

The main conclusions are as follows.

- (i) If f has (T) and g has (Nil), then $f \boxtimes g$ is persistent (Theorem 3.3).
- (ii) If f has (T) and its certificate factors through one fixed linear form in all polarization slots, then $f \boxtimes g$ is persistent for *every* persistent g (Theorem 3.5).
- (iii) (TR)+(JC) imply (T). The only external input is Jacobson's generalized Engel theorem for weakly closed sets of nilpotent transformations (Theorem 4.2).

- (iv) Consequently, Kronecker closure is unconditional when one factor has dimension $d \leq 3$, for arbitrary degree; for $n = 3$ it is unconditional when one factor has $d \leq 4$ (Corollaries 5.1 and 5.2).
- (v) A fixed-linear-form certificate in all degrees reduces, for each fixed d , to a quartic rank-one problem (Theorem 7.2).

The general theorem of [1] is therefore not established by the arguments in this note, but its unresolved content is considerably more structured than the original Segre-to-global step suggests.

2. THE SEGRE OBSTRUCTION IN THE POLARIZED PROOF

We use the conventions of [1]. If $f \in \text{Sym}^n V$ and $g \in \text{Sym}^n W$, then for decomposable polarization vectors

$$U^{(i)} = v^{(i)} \otimes w^{(i)}, \quad 1 \leq i \leq r = n - 2,$$

the differentiation identities of [1, Proposition 7 and Corollary 8] give

$$(2) \quad H_{f \boxtimes g}(U^{(1)}, \dots, U^{(r)}) = \frac{1}{n!} H_f(v^{(1)}, \dots, v^{(r)}) \boxtimes H_g(w^{(1)}, \dots, w^{(r)}),$$

where the right-hand side is now an ordinary matrix Kronecker product. Therefore, for persistent f and g ,

$$(3) \quad \det H_{f \boxtimes g}(U^{(1)}, \dots, U^{(r)}) = \left[\frac{1}{n!} P_f(v^{(1)}, \dots, v^{(r)}) P_g(w^{(1)}, \dots, w^{(r)}) \right]^{d_1 d_2}.$$

The product inside brackets is bilinear in each pair $(v^{(i)}, w^{(i)})$, so the universal property of the tensor product does indeed define a unique multilinear form

$$(4) \quad P_{f \boxtimes g} : (V \otimes W)^r \longrightarrow \mathbb{C}$$

whose value on decomposable tuples is the bracketed expression in (3).

The issue is that the universal property extends the *candidate root* $P_{f \boxtimes g}$, not the determinant identity. For each polarization slot, $\det H_{f \boxtimes g}$ has degree $D = d_1 d_2$ in the coordinates of $U^{(i)}$, whereas $P_{f \boxtimes g}^D$ has the same degree. Equality of these polynomials on rank-one tensors does not determine them on $V \otimes W$. Their difference may lie in the ideal of the Segre variety.

This is exactly the obstruction noted immediately before Proposition 10 in [1, Remark 9]: polynomial equality on

$$\text{Seg}(\mathbb{P}V \times \mathbb{P}W) \subset \mathbb{P}(V \otimes W)$$

does not imply global polynomial equality because the Segre ideal, generated by the 2×2 minors of the coordinate matrix, is nonzero. The same issue persists independently in each polarization slot.

Representation-theoretically, restriction to decomposable tensors detects the Cartan component

$$\text{Sym}^D V^* \otimes \text{Sym}^D W^* \subset \text{Sym}^D(V \otimes W)^*,$$

while the other Cauchy summands vanish upon the corresponding Segre restriction. Thus a repair must explain why the determinant residual has no

component in this kernel; multilinearity of the proposed root alone does not do so.

Remark 2.1. *This observation does not imply that the polarized identity is false. In fact, the symbolic calculations described in Section 8 repeatedly confirm the identity at nondecomposable polarization tensors. The point is only that an additional structural argument is required.*

3. NORMALIZED POLARIZED HESSIAN SPACES

Let $f \in \text{Sym}^n V$ be persistent, $d = \dim V$, and $r = n - 2$. Write

$$H_f(A) = H_f(a_1, \dots, a_r)$$

for the Hessian matrix of the quadratic partial polarization $\partial_{a_1} \cdots \partial_{a_r} f$. Fix once and for all a certificate P_f satisfying (1).

Definition 3.1 (Admissible base and normalized Hessian space). *An admissible base tuple is $B = (b_1, \dots, b_r) \in V^r$ with $P_f(B) \neq 0$. Set*

$$G_f = H_f(B), \quad M_f(A) = G_f^{-1} H_f(A), \quad \lambda_f(A) = P_f(A)/P_f(B),$$

and

$$(5) \quad N_f(A) = M_f(A) - \lambda_f(A)I.$$

The associated normalized Hessian space is

$$\mathcal{N}_f(B) = \text{Span}\{N_f(A) : A \in V^r\}.$$

Both $H_f(A)$ and $P_f(A)$ are multilinear in the r polarization slots, so $N_f(A)$ is multilinear as well. In particular, relative to a basis of V , $\mathcal{N}_f(B)$ is spanned by at most d^r basis-tuple evaluations.

For later use we record the multilinear extension of (2).

Lemma 3.2 (Global matrix expansion). *Let $h = f \boxtimes g$. For arbitrary $U^{(i)} \in V \otimes W$, choose finite decompositions*

$$U^{(i)} = \sum_{a_i} v_{a_i}^{(i)} \otimes w_{a_i}^{(i)}.$$

For a multi-index $\alpha = (a_1, \dots, a_r)$ put

$$A_\alpha = (v_{a_1}^{(1)}, \dots, v_{a_r}^{(r)}), \quad C_\alpha = (w_{a_1}^{(1)}, \dots, w_{a_r}^{(r)}).$$

Then

$$(6) \quad H_h(U^{(1)}, \dots, U^{(r)}) = \frac{1}{n!} \sum_{\alpha} H_f(A_\alpha) \boxtimes H_g(C_\alpha).$$

Proof. Equation (2) holds on a decomposable choice in each slot. Both sides of (6) are multilinear separately in each $U^{(i)}$, so expanding each chosen decomposition gives the result. Notice that no determinant is taken in this step; this is precisely why the extension is legitimate. $\square$

We now give the first replacement for the disputed Segre-to-global step.

Theorem 3.3 (Triangularizable–nil criterion). *Let $f \in \text{Sym}^n V$ and $g \in \text{Sym}^n W$ be persistent. Suppose there are admissible base tuples B and C such that*

- (a) $\mathcal{N}_f(B)$ is simultaneously strictly triangularizable, and
- (b) every element of $\mathcal{N}_g(C)$ is nilpotent.

Then $f \boxtimes g$ is persistent.

Proof. Write $d_1 = \dim V$, $d_2 = \dim W$, $D = d_1 d_2$, and choose arbitrary polarization tensors $U^{(1)}, \dots, U^{(r)} \in V \otimes W$. Using Lemma 3.2 and factoring the base Hessians gives

$$(7) \quad H_{f \boxtimes g}(U) = \frac{1}{n!} (G_f \boxtimes G_g) \Xi(U),$$

where

$$\Xi(U) = \sum_{\alpha} M_f(A_{\alpha}) \boxtimes M_g(C_{\alpha}).$$

Choose a basis of V in which every $N_f(A)$ is strictly upper triangular. Since $M_f(A) = \lambda_f(A)I + N_f(A)$, the matrix $\Xi(U)$ is block upper triangular, with d_1 identical diagonal blocks

$$(8) \quad Y = \sum_{\alpha} \lambda_f(A_{\alpha}) M_g(C_{\alpha}).$$

Consequently

$$\det \Xi(U) = \det(Y)^{d_1}.$$

Write

$$\mu = \sum_{\alpha} \lambda_f(A_{\alpha}) \lambda_g(C_{\alpha}).$$

Then

$$Y = \mu I + \sum_{\alpha} \lambda_f(A_{\alpha}) N_g(C_{\alpha}) = \mu I + \tilde{N},$$

with $\tilde{N} \in \mathcal{N}_g(C)$. By hypothesis $\tilde{N}$ is nilpotent, so $\det Y = \mu^{d_2}$ and therefore

$$(9) \quad \det \Xi(U) = \mu^D.$$

The multilinear root (4) satisfies, after expansion of the same decompositions,

$$P_{f \boxtimes g}(U) = \frac{1}{n!} \sum_{\alpha} P_f(A_{\alpha}) P_g(C_{\alpha}) = \frac{P_f(B) P_g(C)}{n!} \mu.$$

Finally,

$$\det(G_f \boxtimes G_g) = \det(G_f)^{d_2} \det(G_g)^{d_1} = (P_f(B) P_g(C))^D.$$

Taking determinants in (7) and using (9) gives

$$\det H_{f \boxtimes g}(U) = P_{f \boxtimes g}(U)^D$$

for arbitrary U . The Hessian characterization of persistence [2, Theorem 2(c)] completes the proof. $\square$

A stronger one-sided conclusion is available when the certificate of the triangularized factor has rank one in its polarization slots.

Definition 3.4 (Fixed-linear-form certificate). *We say that f satisfies (a-multi) if*

$$(10) \quad P_f(a_1, \dots, a_r) = c \ell(a_1) \cdots \ell(a_r)$$

for some $0 \neq c \in \mathbb{C}$ and $0 \neq \ell \in V^*$.

When P_f is symmetric, this is the polarized form of condition (a) in [2], namely that the ordinary Hessian is a nonzero scalar times a $d(n-2)$ -th power of a linear form.

Theorem 3.5 (Factorized-certificate criterion). *Let $f \in \text{Sym}^n V$ and $g \in \text{Sym}^n W$ be persistent. Suppose f satisfies (a-multi) and, at an admissible base B , $\mathcal{N}_f(B)$ is simultaneously strictly triangularizable. Then $f \boxtimes g$ is persistent, with no nil-space or triangularizability hypothesis on g .*

Proof. Use the notation of the proof of Theorem 3.3. Since (a-multi) holds and $B = (b_1, \dots, b_r)$ is admissible, define slotwise normalized forms

$$\widehat{\ell}_i(v) = \frac{\ell(v)}{\ell(b_i)}.$$

For decompositions $U^{(i)} = \sum_{a_i} v_{a_i}^{(i)} \otimes w_{a_i}^{(i)}$, set

$$(11) \quad W^{(i)} = \sum_{a_i} \widehat{\ell}_i(v_{a_i}^{(i)}) w_{a_i}^{(i)} \in W.$$

The diagonal block (8) becomes, by multilinearity of M_g ,

$$Y = M_g(W^{(1)}, \dots, W^{(r)}).$$

Persistence of g therefore gives

$$\det Y = \lambda_g(W^{(1)}, \dots, W^{(r)})^{d_2}.$$

As in Theorem 3.3, block triangularity yields

$$\det \Xi(U) = \lambda_g(W^{(1)}, \dots, W^{(r)})^D.$$

On the other hand, the multilinear extension defining $P_{f \boxtimes g}$ gives

$$P_{f \boxtimes g}(U) = \frac{P_f(B)P_g(C)}{n!} \lambda_g(W^{(1)}, \dots, W^{(r)}).$$

The same determinant calculation as before proves the required D -th power identity globally. $\square$

4. JORDAN CLOSURE AND JACOBSON'S GENERALIZED ENGEL THEOREM

The simultaneous triangularizability hypothesis in the preceding theorems is a normal-form condition. The following criterion replaces it with polynomial membership identities in the normalized Hessian space.

Proposition 4.1 (Trace plus Jordan closure implies nilpotence). *Let $\mathcal{N} \subseteq \text{End}(V)$ be a linear space over a field of characteristic zero. Assume*

$$\text{tr } X = 0 \quad \text{for all } X \in \mathcal{N}, \quad XY + YX \in \mathcal{N} \quad \text{for all } X, Y \in \mathcal{N}.$$

Then every $X \in \mathcal{N}$ is nilpotent.

Proof. Jordan closure is equivalent to closure under squaring, since

$$XY + YX = (X + Y)^2 - X^2 - Y^2.$$

Hence $X^k \in \mathcal{N}$ for every $k \geq 1$: from $X^k \in \mathcal{N}$ we obtain $2X^{k+1} = XX^k + X^kX \in \mathcal{N}$. Therefore $\text{tr}(X^k) = 0$ for all $k \geq 1$. Newton's identities imply that all non-leading coefficients of the characteristic polynomial of X vanish, so $\chi_X(t) = t^{\dim V}$. Thus X is nilpotent. $\square$

The remaining passage from pointwise nilpotence to simultaneous strict triangularizability is false for arbitrary linear spaces of matrices. Jordan closure supplies exactly the extra multiplicative closure needed for a classical theorem of Jacobson.

Theorem 4.2 (Trace–Jordan criterion). *Let $\mathcal{N} \subseteq \text{End}(V)$ be a finite-dimensional linear space over $\mathbb{C}$. If $\mathcal{N}$ is trace zero and Jordan closed, then $\mathcal{N}$ is simultaneously strictly triangularizable.*

Proof. By Proposition 4.1, every element of $\mathcal{N}$ is nilpotent. Moreover $\mathcal{N}$ is a weakly closed set in Jacobson's sense: for every $X, Y \in \mathcal{N}$,

$$XY + 1 \cdot YX \in \mathcal{N}.$$

Jacobson's generalized Engel theorem [4, 5] states that the associative enveloping algebra $\mathcal{A}$ generated by a weakly closed set of nilpotent linear transformations on a finite-dimensional vector space is nilpotent. Thus $\mathcal{A}^m = 0$ for some m .

For completeness, a nilpotent associative algebra of transformations is simultaneously strictly triangularizable. If $\mathcal{A} \neq 0$, choose the largest k with $\mathcal{A}^k V \neq 0$; then any nonzero $v \in \mathcal{A}^k V$ satisfies $\mathcal{A}v = 0$. The induced algebra on $V/\mathbb{C}v$ is again nilpotent. Induction on $\dim V$ gives a complete invariant flag on which every element of $\mathcal{A}$, hence every element of $\mathcal{N}$, acts strictly triangularly. $\square$

Combining Theorems 3.3 and 4.2 gives a theorem-shaped strengthening of the computational Jordan test.

Corollary 4.3. *Let f and g be persistent. If, for some admissible bases, $\mathcal{N}_f$ satisfies $(TR)+(JC)$ and $\mathcal{N}_g$ satisfies (Nil) , then $f \boxtimes g$ is persistent. If f additionally satisfies $(a\text{-multi})$, the hypothesis on g is unnecessary.*

This isolates the following identity-shaped open question.

Question 4.4 (Normalized Hessian Jordan closure). *Does every persistent symmetric tensor admit an admissible base for which its normalized Hessian space is trace zero and Jordan closed?*

A positive answer would imply the triangularizability mechanism needed above. It is stronger than simultaneous triangularizability as a statement about abstract matrix spaces, but it is also more algebraic: Jordan closure can be expressed by explicit polynomial span-membership identities.

5. UNCONDITIONAL LOW-DIMENSIONAL CONSEQUENCES

The classifications in [2, Theorem 5] make the preceding criteria unconditional in several nontrivial regimes.

Corollary 5.1 (One factor of dimension at most three). *Let $f \in \text{Sym}^n \mathbb{C}^{d_1}$ and $g \in \text{Sym}^n \mathbb{C}^{d_2}$ be persistent. If $\min(d_1, d_2) \leq 3$, then $f \boxtimes g$ is persistent.*

Proof. For $d = 2$ and $d = 3$, [2, Theorem 5] gives, up to GL, respectively the normal forms

$$x_0^{n-1}x_1, \quad x_0^{n-1}x_2 + x_0^{n-2}x_1^2.$$

Their Hessian certificates satisfy (a-multi). At the diagonal base $(e_0, \dots, e_0)$, a direct calculation gives

$$\mathcal{N}_{x_0^{n-1}x_1} = \mathbb{C}E_{21}$$

and, for the ternary normal form,

$$\mathcal{N} \subseteq \text{Span}\{E_{21} + c_n E_{32}, E_{31}\}$$

for a nonzero scalar c_n . Both spaces are strictly lower triangular. Thus the low-dimensional factor satisfies the hypotheses of Theorem 3.5; symmetry of the Kronecker product in its two factors finishes the proof. $\square$

For cubics there is a further dimension-four case.

Corollary 5.2 (Cubics with one factor of dimension at most four). *Let $f \in \text{Sym}^3 \mathbb{C}^{d_1}$ and $g \in \text{Sym}^3 \mathbb{C}^{d_2}$ be persistent. If $\min(d_1, d_2) \leq 4$, then $f \boxtimes g$ is persistent.*

Proof. For $n = 3$, the certificate has one slot, so (a-multi) is automatic. The cases $d \leq 3$ follow from Corollary 5.1. For $d = 4$, [2, Theorem 5] puts a persistent cubic, up to GL, in the form

$$f = ax_0^2x_3 + bx_0x_1x_2 + cx_1^3, \quad ab \neq 0.$$

Its Hessian at e_0 is nonsingular, and with $M(v) = H_f(e_0)^{-1}H_f(v)$ one computes

$$M(v) - v_0I = \begin{pmatrix} 0 & 0 & 0 & 0 \\ v_1 & 0 & 0 & 0 \\ v_2 & \frac{6c}{b}v_1 & 0 & 0 \\ v_3 & \frac{b}{2a}v_2 & \frac{b}{2a}v_1 & 0 \end{pmatrix}.$$

Hence the normalized Hessian space is simultaneously strictly lower triangular. Apply Theorem 3.5. $\square$

These corollaries are independent of the disputed Segre-to-global inference in [1, Proposition 10].

6. A CUBIC ALGEBRA INTERPRETATION

For cubics the Jordan-closure condition has an especially rigid interpretation. Let $f \in \text{Sym}^3 V$ be persistent, choose $b \in V$ with $H_f(b)$ nonsingular, and set

$$G = H_f(b), \quad M(u) = G^{-1}H_f(u).$$

Then $M(b) = I$. Full symmetry of the cubic tensor implies

$$(12) \quad M(u)v = M(v)u \quad (u, v \in V).$$

Indeed, pairing both sides with an arbitrary w using the nondegenerate form G reduces (12) to symmetry of the associated trilinear form. Thus

$$u * v := M(u)v$$

defines a commutative bilinear product on V with unit b .

Proposition 6.1. *Assume the cubic normalized Hessian space is Jordan closed. Then the product $*$ above is associative. Equivalently, $M(V)$ is a unital commutative associative subalgebra of $\text{End}(V)$ and $M(u)$ is left multiplication by u .*

Proof. Jordan closure of $\mathcal{N}_f$ is equivalent to Jordan closure of $M(V) = \mathbb{C}I + \mathcal{N}_f$. Since persistence implies conciseness by definition [2, Definition 1], the linear map $M : V \rightarrow \text{End}(V)$ is injective. Hence for $u, v \in V$ there is a unique element $q(u, v)$ with

$$M(q(u, v)) = \frac{1}{2}(M(u)M(v) + M(v)M(u)).$$

Apply both sides to the unit b . By (12), $q(u, v) = u * v$, so

$$(13) \quad M(u * v) = \frac{1}{2}(M(u)M(v) + M(v)M(u)).$$

Applying (13) to w gives

$$2(u * v) * w = u * (v * w) + v * (u * w).$$

Cyclically permuting u, v, w yields three equations. Subtracting any two shows $(u * v) * w = u * (v * w)$ (over characteristic zero), hence associativity. $\square$

This is slightly stronger than merely obtaining a Jordan algebra: in the Hessian setting, Jordan closure of the regular multiplication operators forces the underlying commutative product to be associative. It suggests studying persistent cubics through finite-dimensional local/Frobenius-type algebras rather than through arbitrary nilpotent matrix spaces.

7. A VERONESE CONTRACTION REDUCTION TO QUARTICS

The factorized-certificate condition (a-multi) is known in the classified low-dimensional cases but is open in general. The following reduction localizes its entire higher-degree content in degree four.

Definition 7.1. *A form $f \in \text{Sym}^m \mathbb{C}^d$, $m \geq 3$, is certified if there is a nonzero multilinear form*

$$P_f : (\mathbb{C}^d)^{m-2} \rightarrow \mathbb{C}$$

such that $\det H_f = P_f^d$ identically in the polarization variables. By [2, Theorem 2(c)], certified and persistent are equivalent in the present setting.

Theorem 7.2 (Veronese contraction lemma). *Fix d . Suppose every certified quartic in $\text{Sym}^4 \mathbb{C}^d$ satisfies (a-multi). Then every certified form in $\text{Sym}^m \mathbb{C}^d$, for every $m \geq 3$, satisfies (a-multi).*

Proof. We induct on m . The case $m = 3$ is immediate because the certificate has one slot; $m = 4$ is the hypothesis. Assume $m \geq 5$, set $r = m - 2 \geq 3$, and write $P = P_f$.

Define the linear contraction map

$$\Phi : V \longrightarrow (V^*)^{\otimes(r-1)}, \quad \Phi(u) = \iota_u P = P(u, -, \dots, -).$$

Its kernel K is a proper linear subspace because $P \neq 0$. For $u \notin K$, commutativity of directional derivatives gives the exact matrix identity

$$H_{\partial_u f}(v_2, \dots, v_r) = H_f(u, v_2, \dots, v_r).$$

Hence $\partial_u f$ is certified and its certificate root is, up to a scalar d -th root of unity, $\Phi(u)$. By induction,

$$(14) \quad \Phi(u) = c_u \ell_u^{\otimes(r-1)}, \quad u \notin K,$$

for some $c_u \neq 0$ and $\ell_u \neq 0$.

We claim that all ℓ_u in (14) are proportional. Otherwise choose $u_1, u_2 \notin K$ with independent ℓ_1, ℓ_2 . The affine line $u_1 + tu_2$ meets the linear subspace K in at most one point, so choose $t \neq 0$ with $u_t = u_1 + tu_2 \notin K$. By linearity of Φ ,

$$c_t \ell_t^{\otimes p} = c_1 \ell_1^{\otimes p} + t c_2 \ell_2^{\otimes p}, \quad p = r - 1 \geq 2.$$

Restricting to the diagonal gives

$$a \ell_1(x)^p + b \ell_2(x)^p = e \ell(x)^p, \quad a b e \neq 0.$$

After absorbing p -th roots of a, b , the left side is $X^p + Y^p$ for independent linear forms X, Y . Over $\mathbb{C}$ this factors into $p \geq 2$ distinct, pairwise nonproportional linear factors, whereas the right side is a pure p -th power. Unique factorization gives a contradiction.

Thus the image of Φ is contained in the one-dimensional space $\mathbb{C} \ell^{\otimes(r-1)}$ for a fixed ℓ . Since $V \setminus K$ spans V , there is a linear form $c \in V^*$ with

$$(15) \quad P(u, v_2, \dots, v_r) = c(u) \ell(v_2) \cdots \ell(v_r)$$

for all arguments.

It remains to identify c with ℓ . The determinant $\det H_f$ is symmetric in the r polarization slots, so for every $\sigma \in S_r$, $(P \circ \sigma)^d = P^d$. Since the polynomial ring is an integral domain, $P \circ \sigma = \chi(\sigma)P$ for a character $\chi : S_r \rightarrow \mu_d$. Hence P is either symmetric or alternating. If it were alternating, then $\iota_u P$ would be alternating in its $r - 1 \geq 2$ remaining slots, while (15) makes it symmetric; all contractions would vanish, contradicting $P \neq 0$. Thus P is symmetric. Swapping the first two slots in (15) gives

$$c(u)\ell(v) = c(v)\ell(u),$$

so $c = \gamma\ell$ for some $\gamma \neq 0$. Therefore $P = \gamma\ell^{\otimes r}$, which is (a-multi). $\square$

Remark 7.3 (The quartic base case). *The proof fails exactly at $r = 2$: then $p = r - 1 = 1$, and the degree-one Veronese contains lines. Moreover the alternating possibility is not eliminated by the last contraction argument. Thus the remaining base problem is genuinely: for a certified quartic, prove that the bilinear certificate P_f is symmetric of rank one (and in particular not alternating). This is not supplied by the current low-dimensional classification for arbitrary $d \geq 4$.*

8. EXACT SYMBOLIC EVIDENCE

The preceding results arose from exact symbolic experiments designed to test possible repairs rather than merely to resample the Segre-restricted identity. All arithmetic was over $\mathbb{Q}$ using SymPy; no floating-point tests were used. For each certified form, an admissible diagonal base was chosen when available, the space $\mathcal{N}_f$ was generated from all basis tuples, reduced to an exact linear basis, and the following gates were checked:

$$(\text{TR}), \quad (\text{JC}), \quad \dim \bigcap_{N \in \mathcal{N}_f} \ker N > 0.$$

Jordan closure was tested by exact rank membership for every unordered pair of a reduced basis of $\mathcal{N}_f$.

The corpus contained:

- ten named normal forms, including the binary W_n family, the ternary normal forms, the classified persistent cubics in dimension four, and the quartic $x_0x_2^3 + x_1x_2^2x_3 + x_3^4$ from [2, Example 28];
- all 156 certificate-passing forms found among the $\binom{35}{1} + \binom{35}{2} + \binom{35}{3} = 7175$ coefficient-one sums of at most three distinct quartic monomials in four variables;
- 151 accumulated certificate-passing quintics in four variables from the analogous partial search; and
- four Kronecker products whose normalized Hessian spaces were constructed directly from the product tensor, including product dimensions 4, 6, 8, and 9.

One named quartic is also present in the 156-form hunt, so these data comprise 321 gate executions but 320 distinct literal polynomial inputs before any GL-orbit reduction. After literal deduplication, 2204 unordered Jordan-pair membership tests remain. No Jordan-closure failure was observed.

The negative controls were informative. Six symmetric but nonpersistent forms failed (JC) in every case. The strictly triangular space $\text{Span}\{E_{12}, E_{23}\}$ shows that simultaneous triangularizability does not imply Jordan closure, since

$$E_{12}E_{23} + E_{23}E_{12} = E_{13} \notin \text{Span}\{E_{12}, E_{23}\}.$$

Conversely, a certificate-satisfying bilinear toy family that is not the polarized Hessian of a symmetric tensor passes (JC) but fails (TR); thus (JC) alone is not sufficient.

A further empirical regularity is

$$(16) \quad \dim \mathcal{N}_f \in \{d-1, d\}$$

for every distinct tested persistent input, even when $r \geq 2$ and the a priori spanning set has size d^r . No proof of (16) is known to us.

There are important sampling limitations. The quartic sweep uses coefficient one only, the quintic block is partial, no $d = 5$ hunt was completed, and the known examples are heavily biased toward the fixed-linear-form condition. The computations should therefore be read as evidence for structural conjectures, not as a substitute for proof. Ancillary scripts and exact outputs will accompany a public version of this note.

9. OPEN PROBLEMS AND STATUS OF KRONECKER CLOSURE

The following problems now appear to isolate the main missing structure.

Question 9.1. *Does persistence imply $(TR)+(JC)$ for a normalized polarized-Hessian space at some admissible base?*

By Theorem 4.2, a positive answer implies simultaneous strict triangularizability and therefore supplies the matrix-space input to the Kronecker repair.

Question 9.2 (Quartic rank-one problem). *If $f \in \text{Sym}^4 \mathbb{C}^d$ is certified and*

$$\det H_f(u, v) = P_f(u, v)^d,$$

must P_f be a symmetric bilinear form of rank one?

A positive answer for each fixed d implies, by Theorem 7.2, the fixed-linear-form condition in every degree in that dimension.

Question 9.3. *For a persistent symmetric tensor, is $\dim \mathcal{N}_f \leq d$? More strongly, under natural nondegeneracy hypotheses, must $\dim \mathcal{N}_f \in \{d-1, d\}$?*

There is also a direct route that bypasses normalized-Hessian triangularization. Let

$$R_{f,g}(U) = \det H_{f \boxtimes g}(U) - P_{f \boxtimes g}(U)^D.$$

The decomposable calculation shows that $R_{f,g}$ lies in the appropriate sum of Segre ideals after restriction in each polarization slot. A direct proof of Kronecker closure could instead show that the persistence/Hessian relations on f and g force every non-Cartan Cauchy component of $R_{f,g}$ to vanish. This is an ideal-membership or syzygy problem rather than a density argument.

To summarize: the proof of [1, Proposition 10] requires an additional global argument, but the general closure theorem remains plausible. The results above establish it in substantial unconditional subcases and replace the Segre-to-global step by explicit algebraic conditions whose general validity can be attacked independently.

A NOTE ON PROCESS

Learned Response Labs is a research collaboration between Adam Guetz and AI systems (primarily ChatGPT and Claude). This note originated in a reading of Gharahi’s preprint [1]; the gap analysis, the repair mathematics, the drafting of this text, and the verification code were primarily generated by AI systems (OpenAI GPT and Anthropic Claude) operating adversarially: results produced by one system were independently rederived and adversarially audited by another before internal acceptance, and every computation is exact rational arithmetic reproducible from the ancillary bundle. Adam designed and directed the research program and made the circulation decision but did not perform mathematical verification; no human has. Accordingly, the results are presented as *proof candidates* that have undergone extensive computational and cross-model audit but no independent human mathematical verification. We invite external review and particularly welcome corrections to any argument or computation.

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The starting point for this note is Masoud Gharahi’s preprint [1] and the Hessian characterization developed with Giorgio Ottaviani in [2]. I am grateful for the clarity of those formulations, which makes the obstruction and the matrix-space alternatives unusually explicit. This is a preliminary note; comments and corrections are welcome.

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